

Path Independence Solutions

1. $\vec{F}$ conservative $\Rightarrow \exists f$ s.t. $\nabla f = \vec{F}$

$$g(x,y,z) = \int_{(1,-2,0)}^{(x,y,z)} \vec{F} \cdot d\vec{r} = f(x,y,z) - f(1,-2,0)$$

$$\begin{aligned} \Rightarrow \nabla g &= \nabla (f(x,y,z) - \overbrace{f(1,-2,0)}^{\text{constant}}) \\ &= \nabla f(x,y,z) - 0 \\ &= \vec{F} \end{aligned}$$

2. (a) $\vec{F} = [2x, 3y, 4z]$

$$\Rightarrow \text{curl}(\vec{F}) = [0-0, 0-0, 0-0] = \vec{0}$$

$$\therefore \boxed{\vec{F} \text{ is conservative}}$$

(b) $\vec{F} = [y, x+z, -y]$

$$\Rightarrow \text{curl}(\vec{F}) = [-1-1, 0-0, 1-1] = [-2, 0, 0] \neq \vec{0}$$

$$\therefore \boxed{\vec{F} \text{ is not conservative}}$$

(c) $\vec{F} = [-y, x, 0]$

$$\Rightarrow \text{curl}(\vec{F}) = [0-0, 0-0, 1-(-1)] = [0, 0, 2] \neq \vec{0}$$

$$\therefore \boxed{\vec{F} \text{ is not conservative}}$$

(d) $\vec{F} = [e^{y+2z}, xe^{y+2z}, 2xe^{y+2z}]$

$$\begin{aligned} \Rightarrow \text{curl}(\vec{F}) &= [2xe^{y+2z} - 2xe^{y+2z}, 2e^{y+2z} - 2e^{y+2z}, e^{y+2z} - e^{y+2z}] \\ &= [0, 0, 0] = \vec{0} \end{aligned}$$

$$\therefore \boxed{\vec{F} \text{ is conservative.}}$$

$$3. \quad \vec{F} = [2x, 3y, 4z] = [f_x, f_y, f_z]$$

$$f_x = 2x$$

$$f = x^2 + g(y, z)$$

$$3y = f_y = 0 + g_y$$

$$g_y = 3y$$

$$g = \frac{3}{2}y^2 + h(z)$$

$$f = x^2 + \frac{3}{2}y^2 + h(z)$$

$$4z = f_z = 0 + h_z$$

$$h_z = 4z$$

$$h = 2z^2 + C$$

$$\therefore \boxed{f(x, y, z) = x^2 + \frac{3}{2}y^2 + 2z^2} \quad (C=0)$$

$$\vec{F} = [e^{y+2z}, xe^{y+2z}, 2xe^{y+2z}] = [f_x, f_y, f_z]$$

$$f_x = e^{y+2z}$$

$$f = xe^{y+2z} + g(y, z)$$

$$xe^{y+2z} = f_y = xe^{y+2z} + g_y$$

$$g_y = 0 \Rightarrow g = h(z)$$

$$f = xe^{y+2z} + h(z)$$

$$2xe^{y+2z} = f_z = 2xe^{y+2z} + h_z$$

$$h_z = 0 \Rightarrow h = C$$

$$\therefore \boxed{f(x, y, z) = xe^{y+2z}} \quad (C=0)$$